72 Lumbar Microdiskectomy

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INDICATIONS

- Patients with back pain, sciatic pain, Lasègue sign (pain with straight leg raise), or sensory deficit that fails to improve with conservative measures.
- Referable foraminal stenosis from posterolateral disk herniation.
- New or progressive motor deficits require more urgent surgical intervention.

CONTRAINDICATIONS

- Large disk herniation causing central canal stenosis requires a laminectomy.
- A more extensive procedure, such as interbody fusion, may be beneficial to patients with recurrent disk herniation or concomitant instability from degenerative disease or spondylolisthesis.

PLANNING AND POSITIONING

- Preoperative evaluation includes a thorough neurologic history and sensory, motor, and reflex examination. The sensory examination should be conducted to rule out any dermatomal distribution of sensory loss. Routine lateral, flexion, and extension lumbar spine films can show dynamic instability that may necessitate additional lumbar fusion. In the event of instability, a diskectomy would treat the radiculopathic leg pain but would not improve mechanical back pain. Magnetic resonance imaging (MRI) without contrast enhancement of the lumbar spine can show focal neural foraminal stenosis and disk herniation and allows selection of the appropriate level for diskectomy and foraminotomy.
- Either prone or lateral position is acceptable, depending on surgeon preference.

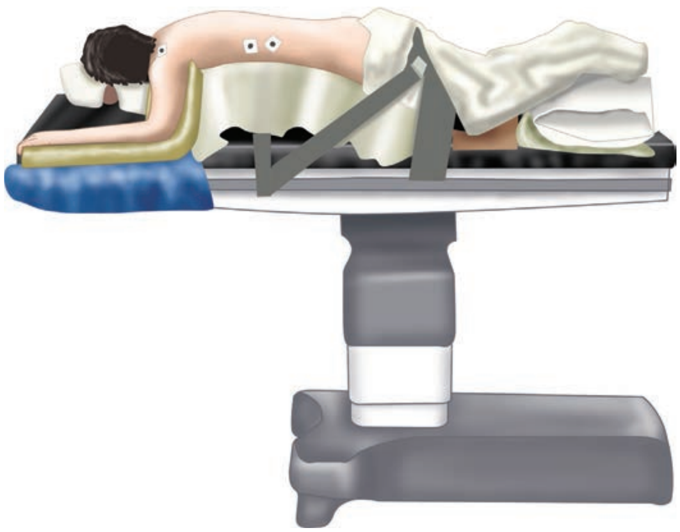

Fig. 72.1 The patient is placed prone on chest rolls or a Wilson frame to hold the spine in flexion.

- The patient should be given a dose of preoperative antibiotics prior to skin incision.

PROCEDURE

Skin Incision

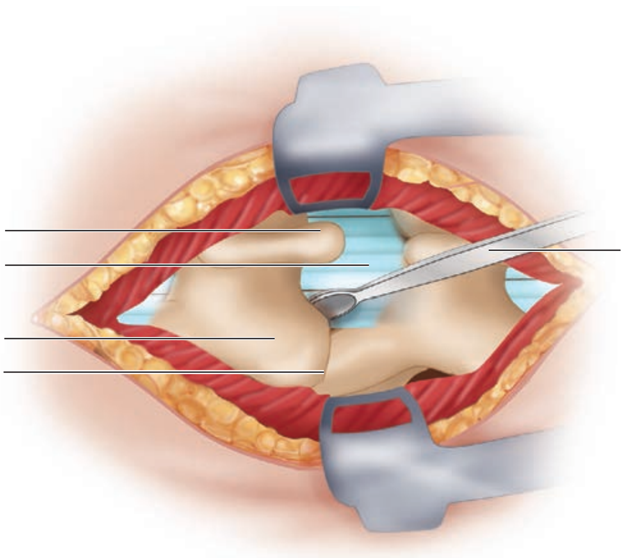

Fig. 72.2 By palpating the anterior superior iliac crest, the L4-5 interspace can be localized for a rough estimation of the level. A spinal needle localization film helps determine the correct site for the exposure. After injection of local anesthetic with epinephrine into the skin and deeper tissues, a small incision just lateral to the midline should be made using a No. 10 blade.

Fascial and Subfascial Dissection

- Bovie electrocautery can be used for subcutaneous dissection and for achieving hemostasis. The subcutaneous fat is dissected until the thoracolumbar fascia is identifiable. A dry sponge raked along the fascia can identify the white tissue easily, and a periosteal elevator can be used to dissect off the subcutaneous fat from the fascia beneath.

Subperiosteal Dissection

- Bovie electrocautery on cut function can be used to incise the fascia. For microdiskectomy, a paramedian facial incision helps avoid damaging midline ligamentous structures. After a minimal, one-sided exposure is completed, a localizing film should be done to confirm the correct level.
- Subperiosteal dissection can also be done rapidly with an open dry sponge raked ventrally and laterally along the spinous process and lamina with a large periosteal dissector, such as a Cobb elevator.

Hemilaminectomy

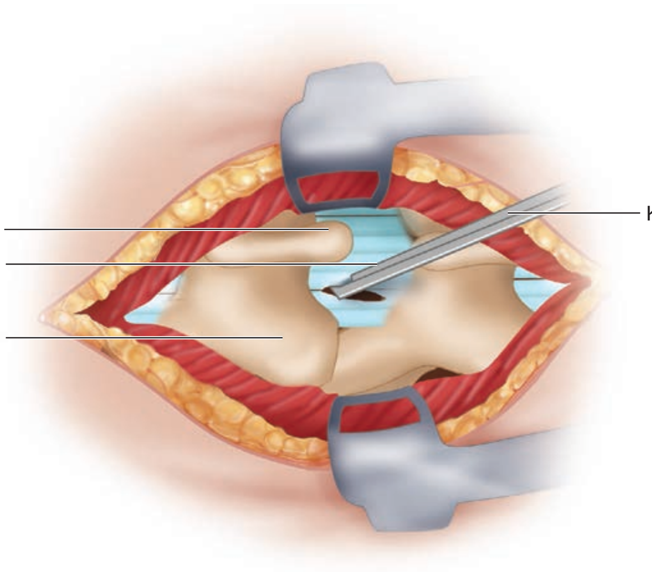

Fig. 72.3 After the correct level has been identified, planning the hemilaminectomy can begin. First, find the soft interspace between two laminae of interest. Thin the bone in both the superior and inferior laminae down through the deep cancellous bone. An up-going curette can be used to remove the remaining bone. A semicircular or square laminectomy can be completed with 1- to 3-mm Kerrison rongeurs and an up-going curette.

Removal of Ligamentum Flavum to Find Thecal Sac

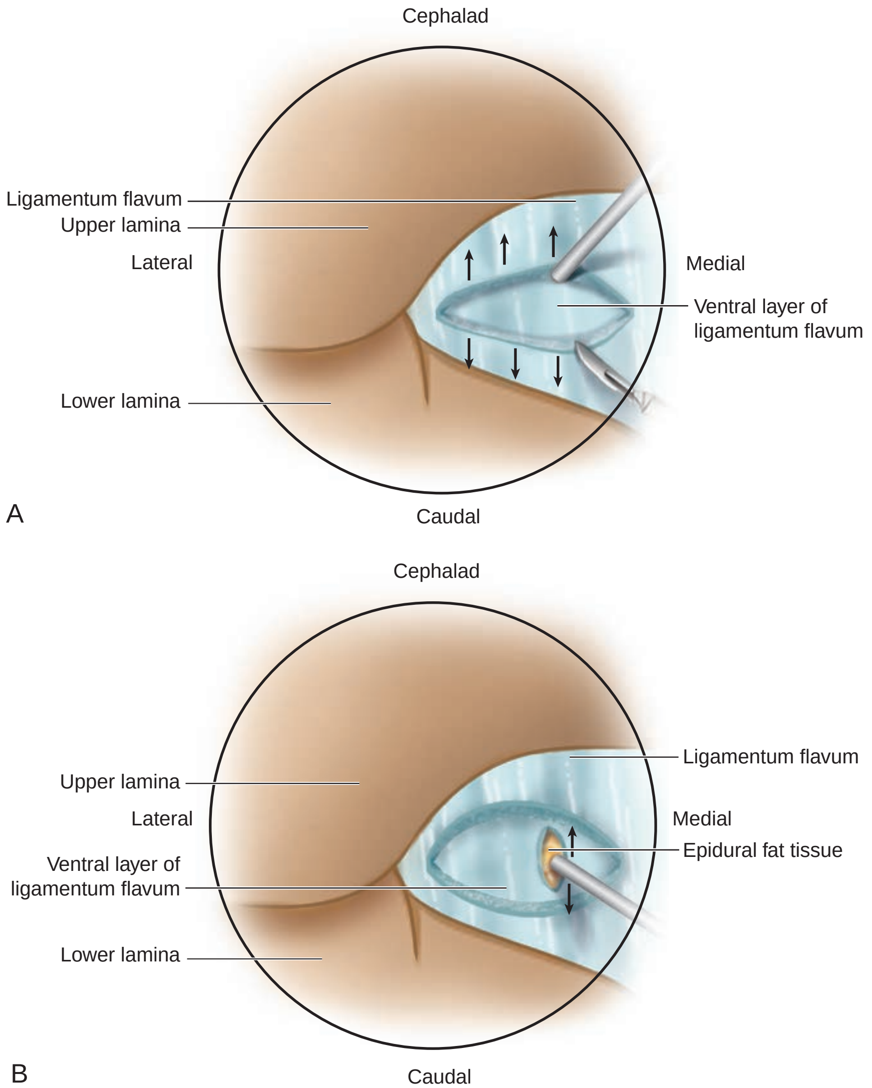

Fig. 72.4 A sharp right-angled instrument can be inserted into the ligamenta flava and used to pull dorsally away from the thecal sac while cutting along the instrument using a No. 15 blade. Dissection can be carefully continued until the whitish-blue thecal sac is identified. Careful removal of the window of ligamenta flava can be done with a pituitary rongeur. The rest of the ligamenta flava is removed with Kerrison rongeurs to create the largest working window possible. The medial segment may be left to help protect the nerve root.

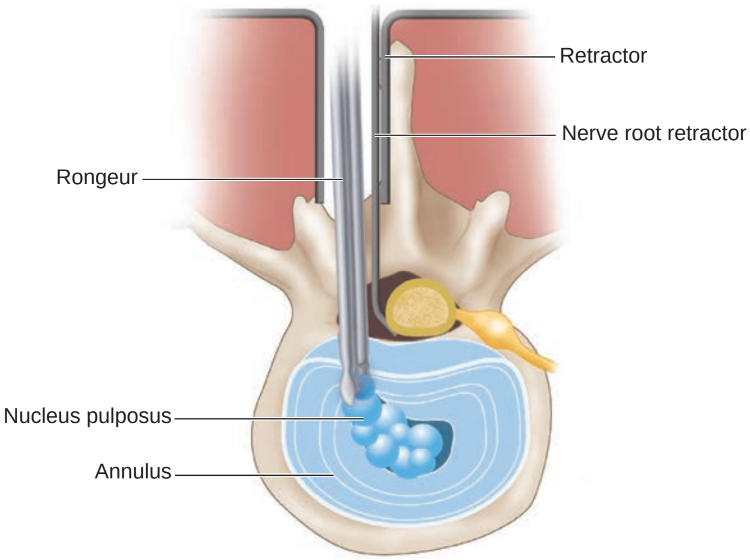

Fig. 72.5 After hemostasis is achieved, countertension-free traction is achieved with a nerve root retractor, and the posterior longitudinal ligament can be carefully opened with a No. 11 blade. The incision is made in a medial to lateral direction to direct the sharp end of the blade away from the dura. A pituitary rongeur can then be used to remove the disk material. A down-going Epstein curette or right-angled Williams instrument can be used to push down paracentral disk material into the now decompressed disk space.

Closure of Operative Site

- Copious irrigation should be used to wash the surgical bed, and any bleeding source should be identified and controlled. Closure should be done meticulously to prevent cerebrospinal fluid leak in the event of a subclinical violation and to allow for a quick and less painful recovery. A watertight fascial closure should be completed with 0-size interrupted, noninverted sutures at narrow 5- to 8-mm intervals.
- Subcutaneous closure is done with 2-0 inverted, interrupted sutures at equally narrow intervals to ensure adequate strength of the closure. The skin is reapproximated and closed with either staples or a running nylon suture.

TIPS FROM THE MASTERS

- Use a straight cupped curette to remove soft tissue and to find the interspinous ligament between the two laminae. When the ligament is visible, resist the temptation to dissect away under the inferior aspect of the superior lamina because this would cause bleeding and ultimately interfere with the hemilaminectomy. Using a 2-mm side-cutting "matchstick" burr or a 3-mm acorn drill bit, drill medially to laterally along the lamina toward the facet joint.
- Control bone bleeding with wax on a Freer elevator applied using a {1/2}×{1/2} cottonoid and bayonetted forceps.
- Cut with the blade away from the dura, and cut from medial to lateral. On the rostral-to-caudal cuts, be careful not to cut too medially.
- Radical versus conservative diskectomy remains an open topic of discussion. Reherniation is a feared complication of the conservative approach, but aggressive resection can produce back pain and risks complications from abdominal injury.

PITFALLS

When drilling, avoid taking more than 50% of the medial facet or entering the facet capsule, as these can destabilize the joint and cause facet joint pain. Similarly, be cautious drilling rostrally, as an iatrogenic pars defect can also lead to instability.

Operating on levels with associated spondylolisthesis may require upfront fusion.

Age, active smoking, and ongoing compensation claims are associated with lower postoperative rates of improvement

BAILOUT OPTIONS

- Dural violation requires primary closure under microscopy. Placement of 6-0 nylon suture using a Castro-Viejo needle driver or long Ryder needle driver may be necessary to achieve closure. Fibrin glue and other liquid sprayable polymers can also be used. Watertight fascial closure helps decrease postoperative cerebrospinal fluid leaks.
- For patients with severe systemic disease, the procedure may be performed under spinal anesthesia.